

The Limit of a Function

Section 1.3

A. Limits

DEFINITION: $\lim_{x \rightarrow a} f(x) = L$ The limit of $f(x)$ as x approaches a , equals L . (Where is the functions value headed as x is “on its way” to a ?)

$\lim_{x \rightarrow a^-} f(x)$ The limit of $f(x)$ as x approaches a from the LEFT.

$\lim_{x \rightarrow a^+} f(x)$ The limit of $f(x)$ as x approaches a from the RIGHT.

B. Techniques of Solving Limits

1. Evaluation

When possible (without violating domain rules) “plug it in.”

Example

1. (Video) $\lim_{x \rightarrow 3} x^2 + 2x - 3 =$

2. (Video) $\lim_{x \rightarrow 1} 5 - \frac{1}{x} =$

3. $\lim_{x \rightarrow 3} x^2 =$

4. $\lim_{x \rightarrow 1} \frac{1}{x} =$

2. Factoring/Manipulation (then Evaluation)

Factor expressions and cancel any common terms.

Example

5. (Video) $\lim_{x \rightarrow 4} \frac{3x - 12}{x^2 - 16} =$

6. (Video) $\lim_{x \rightarrow 3} \frac{x^2 - 3x}{x^2 - 5x + 6} =$

7. $\lim_{x \rightarrow 4} \frac{x-4}{x^2-16} =$

8. $\lim_{x \rightarrow 3} \frac{x^2-3x}{x^2-2x-3} =$

3. Table

Set up a table as x approaches the limit from the left and from the right.

Example

9. (Video) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} =$

10. $\lim_{x \rightarrow 0} \frac{\sin x}{x} =$

$x \rightarrow 0^-$	-0.1	-0.01	-0.001
$\frac{\sin x}{x}$			

$x \rightarrow 0^+$	0.1	0.01	0.001
$\frac{\sin x}{x}$			

IDENTITY: $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

4. Graphing

Graph the function and inspect. (Warning: Your graphing calculator might not always indicate a hole or small discontinuity in a graph. Be sure to always check the domain for restrictions.)

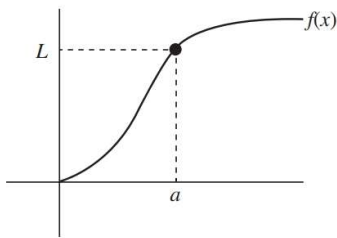
Example

11. $\lim_{x \rightarrow 0} \frac{\sin x}{x} =$

12. $\lim_{x \rightarrow 0^+} \frac{1}{x} =$

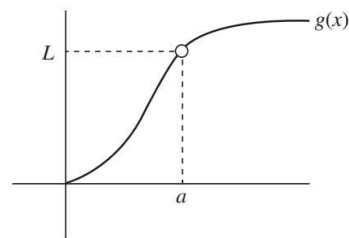
More Examples

13.



$$f(a) =$$

$$\lim_{x \rightarrow a} f(x) =$$



$$g(a) =$$

$$\lim_{x \rightarrow a} g(x) =$$

14.

$$f(x) = \begin{cases} 7-x & \text{if } x \leq -4 \\ x & \text{if } -4 < x \leq 2 \\ (x-1)^2 & \text{if } x > 2 \end{cases}$$

$$\lim_{x \rightarrow -4^-} f(x)$$

$$\lim_{x \rightarrow 2^-} f(x)$$

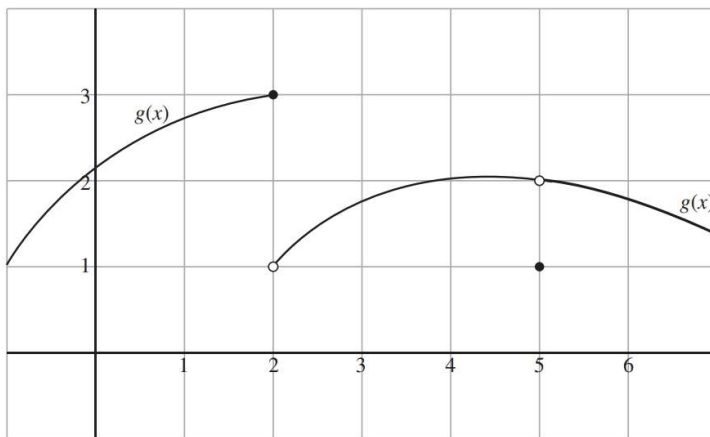
$$\lim_{x \rightarrow -4^+} f(x)$$

$$\lim_{x \rightarrow 2^+} f(x)$$

$$\lim_{x \rightarrow -4} f(x)$$

$$\lim_{x \rightarrow 2} f(x)$$

15.



$$g(2) =$$

$$g(5) =$$

$$\lim_{x \rightarrow 2^-} g(x) =$$

$$\lim_{x \rightarrow 5^-} g(x) =$$

$$\lim_{x \rightarrow 2^+} g(x) =$$

$$\lim_{x \rightarrow 5^+} g(x) =$$

$$\lim_{x \rightarrow 2} g(x) =$$

$$\lim_{x \rightarrow 5} g(x) =$$

C. Average Velocity

DEFINITION: $Velocity = \frac{Distance}{Time}$

Example

16. A ball is thrown straight up into the air at an initial velocity of 75 ft/sec, its height in feet t seconds is given by $y = 75t - 16t^2$.

a. Find the average velocity for the period beginning when $t = 2$ and lasting

i. 0.1 seconds (i.e., the time period $[2, 2.1]$)

ii. 0.01 seconds

iii. 0.001 seconds

b. Estimate the instantaneous velocity of the ball when $t = 2$.